

Literature Status for arXiv:1302.4618: The $4M - 4$ Conjecture

Run fa_banach_001, agent_lane_03

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Status

This is a `literature_already_answered` packet. It records that the $4M - 4$ conjecture posed by Bandeira–Cahill–Mixon–Nelson in *Saving phase: Injectivity and stability for phase retrieval*, arXiv:1302.4618, has already been settled in later literature: the generic sufficiency half was proved by Conca–Edidin–Hering–Vinzant [2], and the proposed necessity half was disproved by Vinzant [3].

Source Question

On PDF page 7 of arXiv:1302.4618, after a dimension-counting discussion of rank-at-most-two Hermitian matrices, the authors pose the following $4M - 4$ conjecture. Let

$$\Phi = \{\varphi_n\}_{n=1}^N \subseteq \mathbb{C}^M, \quad \mathcal{A} : \mathbb{C}^M / \mathbb{T} \rightarrow \mathbb{R}^N, \quad (\mathcal{A}(x))(n) = |\langle x, \varphi_n \rangle|^2.$$

If $M \geq 2$, then:

- (a) if $N < 4M - 4$, then $\mathcal{A}$ is not injective;
- (b) if $N \geq 4M - 4$, then $\mathcal{A}$ is injective for generic Φ .

The source paper proves this conjecture only for $M = 2$ and $M = 3$.

Supporting Answers

The first later paper, arXiv:1312.0158, explicitly identifies the same conjecture. Its Theorem 1.1 states that if $N \geq 4M - 4$, then for a generic frame Φ , the map $\mathcal{A}_\Phi$ is injective. The introduction states that this proves part (b) of the $4M - 4$ conjecture from [1]. Thus the generic sufficiency half of the source conjecture is answered positively.

The second later paper, arXiv:1502.04656, explicitly gives a counterexample to the proposed necessity half. Its abstract presents a complex frame of eleven vectors in four-dimensional complex space and proves that it defines injective measurements; the same paragraph says this disproves the conjecture of Bandeira–Cahill–Mixon–Nelson. In Section II the paper displays the eleven-vector frame in $\mathbb{C}^4$ and states that it provides a counterexample to Conjecture 2 of [1]. Since $4M - 4 = 12$ when $M = 4$, this is an injective example with $N = 11 < 12$, contradicting part (a).

Scope

Together these papers settle the $4M - 4$ conjecture as stated in arXiv:1302.4618: part (b) is true, while part (a) is false. This packet does not claim a complete classification of the exact minimum number of complex measurements in all dimensions. It also does not address the source paper’s separate stability questions, quantitative Lipschitz bounds, or refinements connected to Wright’s conjecture.

Search Evidence

The lane-3 queue selected arXiv:1302.4618. The cheap run indexes were searched for the arXiv id, title words, “phase retrieval”, “4M-4”, and nearby author names; no duplicate packet for this source was found. Targeted searches for phrases including “4M-4 Conjecture phase retrieval”, “Saving phase 4M-4 conjecture”, “generic 4M-4 phase retrieval Conca Edidin Hering Vinzant”, and “small frame certificate injectivity 4M-4” identified arXiv:1312.0158 and arXiv:1502.04656 as the decisive later sources.

References

- [1] A. S. Bandeira, J. Cahill, D. G. Mixon, and A. A. Nelson, *Saving phase: Injectivity and stability for phase retrieval*, arXiv:1302.4618.
- [2] A. Conca, D. Edidin, M. Hering, and C. Vinzant, *An algebraic characterization of injectivity in phase retrieval*, arXiv:1312.0158.
- [3] C. Vinzant, *A small frame and a certificate of its injectivity*, arXiv:1502.04656.